Cian Hughes

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GitHub: github.com/Cian-H | **GitLab:** gitlab.com/Cian-H | **Gitea:** gitea.bulba.space | **Blog:** blog.bulba.space | **Google Scholar:** scholar.google.com/citations?user=Rn6JTuAAAAAJ | **Semantic Scholar:** www.semanticscholar.org/author/Cian-Hughes/1706865947 | **ResearchGate:** www.researchgate.net/profile/Cian-Hughes?ev=hdr_xprf | **LinkedIn:** linkedin.com/in/cian-hughes-03211183

PROFILE

A versatile AI researcher and High-Performance Computing (HPC) engineer with over 10 years of experience bridging the gap between advanced scientific research and scalable industrial infrastructure. Holding a PhD in mechanical engineering with a specialization in Neuro-Symbolic AI, I have a proven track record of architecting distributed data systems, deploying machine learning models, and driving algorithmic innovation across extensive published research. I bring a rare blend of interdisciplinary expertise, from leading institutional open-science initiatives and data management committees, to tackling massive datasets in fast-paced collaborative environments. My goal is to apply this unique background to solve complex technical challenges and modernize data ecosystems.

SKILLS

Technical

- 10+ yrs high performance programming experience (Python, Rust, Fortran, Julia, Elixir)
- Experience developing ML models (PyTorch, Keras/Tensorflow, XGBoost)
- Experience with large, quantized, sparse datasets in mixed hardware settings (e.g: CPUs, GPUs, TPUs, FPGAs)
- Project lead for high-throughput data processing libraries using vector processing, GPGPU, SIMD
- Administrator of data repository servers: Invenio deployments via docker and kubernetes on AWS
- Proficient in distributed Big Data processing frameworks and ecosystem tooling (Hadoop, Apache Spark, Hive, Pig, Storm)
- Well versed in debugging, benchmarking, profiling, program optimisation (pdb, snoop, timeit, criterion, godbolt, cachegrind, massif)
- Expertise in declarative infrastructure, reproducible OS configurations, and automated research deployments (Nix, NixOS, CI/CD Pipelines)
- Experience engineering Generative AI integrations (Local LLMs, OpenWebUI, RAG) and researching Neuro-Symbolic Artificial Intelligence architectures
- Competent in full-stack web development and native UI design, building reactive interfaces for scientific projects (Svelte, QML)
- Domain expertise in Additive Manufacturing and Image Processing, seamlessly linking physical system behavior with advanced algorithmic analysis
- Deep understanding of language internals, metaprogramming, and functional programming paradigms (Clojure, Lisp/Scheme, Python ASTs)
- Experience with cluster architecture design and deployment for HPC environments (Slurm-on-Kubernetes, Ceph distributed storage, and integration with DCU SoC's prototype compute architecture).

Personal

- Ability to work collaboratively in team, project management activities, e.g.: development of data science libraries
- Experienced in management of PhD students, masters students, research assistants
- Excellent critical thinking and problem solving ability
- Written/oral/language skills, demonstrated by publication, teaching (e.g. development of university course materials for HPC CS lab), and active technical blogging.
- Strong in interdisciplinary engineering, sciences, and mathematical analysis
- Experienced at problem solving and self learning
- Proven ability to translate complex HPC and machine learning concepts to non-technical stakeholders, management, and domain experts.
- Experienced at distilling open-ended scientific research questions into actionable, scoped technical roadmaps and software architectures.
- Strong advocate for open-science, reproducible research pipelines, and open-source software development.
- Experienced in authoring technical proposals, grant applications, and strategic architectural documentation.

EMPLOYMENT EXPERIENCE

Dublin City University, Dublin, Ireland
Postdoctoral Researcher, October 2023 – Present

- Researching as part of I-Form research centre, applying various ML methods to metal 3d printing datasets
- Primary focus is currently on model architecture development and semantic knowledge injection methodologies in the field of symbolic neural networks, i.e: the development of ways to allow human experts to guide and constrain models for more efficient training
- Currently responsible for hiring and managing a PhD student in collaboration with the primary investigator on my project
- Applying a variety of machine learning methods to train models on extremely large datasets (>50GB per sample), using methods such as quantisation, sparsification, data distillation, and data streaming/pipelining to facilitate this
- Designed and taught a 4th-year university module on distributed computing

Dublin City University, Dublin, Ireland
Postdoctoral Researcher, October 2021 – October 2023

- Research as part of a joint Fraunhofer Institute and I-Form research centre project, on the topic of "Novel Fabrication Methods for Protein Digestion Based Biosensors"
- Developed a new method for fabricating biosensor electrodes to be used in protein digestion based sensors
- Designed method using metal 3d printing and Electrochemical Impedance Spectroscopy (EIS) as basis for a biosensor
- Further developed internal libraries for data handling/analysis

Dublin City University, Dublin, Ireland
Postdoctoral Researcher, August 2020 – October 2021

- Lead Researcher as part of the I-Form research centre, on the topic of "Improved Methodology for Closed Loop Control in AM"
- Responsible for development of data acquisition/handling tools for laser powder bed fusion based metal 3d printing
- Tested various in-situ methods such as ultrasonic monitoring
- Implemented internal libraries for big data handling, analysis and computations on HPC cluster. Built on top of libraries such as dask, using optimised rust code via FFI bindings.

Dublin City University, Dublin, Ireland
Postgraduate Research Assistant, January 2020 – August 2020

- Project Manager as part of the EU Erasmus+ iCARE project
- Responsible for managing project progress on work packages and ensuring continued progress
- Responsible for facilitating the dissemination of research and expertise as part of the project

Dublin City University, Dublin, Ireland
Postgraduate Research Assistant, November 2019 – April 2020

- Early stages of project intended to develop way to plug in and apply machine learning methods to university teaching tools such as Moodle
- Early steps included the setup of a Moodle test instance, decisions on which tools to utilise, and basic experiments analysing course data
- Collaboration with research group who had developed other Moodle plugins using WxMaxima
- However, project was stopped due to disruptions to teaching at beginning of the Covid-19 pandemic

Dublin City University, Dublin, Ireland
Laboratory Demonstrator & Tutor, July 2015 – December 2019

- Responsible for demonstrating experimental setups
- Responsible for teaching students to develop experimental methods and use theoretical concepts in applied setting
- Responsible for marking and grading student submissions and examinations
- Taught numerous courses during this time period, including Classical Mechanisms, Pneumatics, Applied Mathematics, and Materials Science and Engineering

Dublin City University, Dublin, Ireland
Research Assistant, October 2014 – April 2015

- Developed a new industrial method for the production of Donepezil (an Alzheimer's medication)
- Primary experimentalist in development/optimisation of catalytic reaction for production of Donepezil
- Responsible for the synthesis of various feedstocks at industrial scale, as well as the final product
- Responsible for the design of an industrially scalable purification method for final products

Dublin City University, Dublin, Ireland
Undergraduate Researcher, February 2014 – May 2014
- Aided in development of new class of drugs (“partial Calix[4]pyrroles”) to treat Multiple Sclerosis
- Contributed to the design and optimisation of lead compounds
- Responsible for designing synthetic pathways/methods for synthesising difficult to produce derivatives
- Responsible for synthesis of various derivatives to be tested as possible drug candidates

University College Dublin, Dublin, Ireland
Research Consultant, January 2020 – April 2020
- Aiding in project management for Fraunhofer Institute’s “BioManu3” project until its premature end due to Covid-19 pandemic
- Responsible for arranging and facilitating collaboration between numerous project centres across Europe
- Responsible for administrative project management roles such as research, drafting and documenting meetings, converting minutes to actionables and milestones, etc.

EDUCATION

2015 – 2019	PhD in Mechanical Engineering, Dublin City University Awarded for "Laser Assisted Production of Nanostructures for Biomedical Sensing Applications"
2010 – 2014	BSc in Chemical and Pharmaceutical Sciences, Dublin City University

PUBLICATIONS

Laser nanostructured gold biosensor for proto-oncogene detection
Scientific Reports, 2023 | DOI | Link

Resultant physical properties of as-built nitinol processed at specific volumetric energy densities and correlation with in-situ melt pool temperatures
Journal of Materials Research and Technology, 2022 | DOI | Link

Laser Powder Bed Fusion of Aluminium Alloy 6061 for Ultra-High Vacuum Applications
Key Engineering Materials, 2022 | DOI | Link

Assessing dependency of part properties on the printing location in laser-powder bed fusion metal additive manufacturing
Materials Today Communications, 2022 | DOI | Link

In-situ sensing, process monitoring and machine control in Laser Powder Bed Fusion: A review
Additive Manufacturing, 2021 | DOI | Link

Laser beam powder bed fusion of nitinol shape memory alloy (SMA)
Journal of Materials Research and Technology, 2021 | DOI | Link

Emerging Technology Trends in Point-of-Care Diagnostics
2020 International Conference on Assistive and Rehabilitation Technologies (iCareTech), 2020 | DOI

Development of a selective inhibitor for Kv1.1 channels prevalent in demyelinated nerves
Bioorganic Chemistry, 2020 | DOI | Link

Electrochemical and Chronoamperometry Assessment of Nano-gold Sensor Surfaces Produced via Novel Laser Fabrication Methods
Journal of Electroanalytical Chemistry, 2020 | DOI

Trace multi-class organic explosives analysis in complex matrices enabled using LEGO®-inspired clickable 3D-printed solid phase extraction block arrays
Journal of Chromatography A, 2020 | DOI | Link

Wearable devices for monitoring work related musculoskeletal and gait disorders
Proceedings - 2020 International Conference on Assistive and Rehabilitation Technologies, iCareTech 2020, 2020 | DOI | Link

Modification of Polymer Substrates with Extreme Ultraviolet - Potential Application in Cancer Cell Identification
Acta Physica Polonica A, 2018 | DOI | Link

Modelling and optimisation of single-step laser-based gold nanostructure deposition with tunable optical properties
Optics & Laser Technology, 2018 | DOI | Link

Agricultural Feedstock Supplemented with Manganese for Biosurfactant Production by Bacillus subtilis
Waste and Biomass Valorization, 2017 | DOI | Link

Pulsed laser deposition of plasmonic nanostructured gold on flexible transparent polymers at atmospheric pressure
Journal of Physics D: Applied Physics, 2017 | DOI | Link

SERVICE & COMMITTEES

Ongoing	**Reviewer**, Materials (Journal, ISSN 1996-1944) Served as a peer reviewer for the open-access academic journal 'Materials'.
2024	**Workshop Organizer**, 2024 AMS-CDT Conference Organized and ran a workshop session at the conference.
Ongoing	**Committee Member**, I-Form Data Management Committee Primary contributor driving the adoption of data transparency and hosted data repositories. Primary developer and maintainer for deploying openly available data repository platforms.
Ongoing	**Committee Member**, I-Form Equity, Diversity, and Inclusion (EDI) Committee Partnered with the Irish Centre for Diversity to foster an inclusive and equitable research culture within the institute.
Ongoing	**EPE Contributor**, I-Form Research Centre Active participant in judging and providing technical support for Education and Public Engagement (EPE) efforts. Developing new EPEs including AI-assisted design 3D printing workshops in collaboration with local community centres.